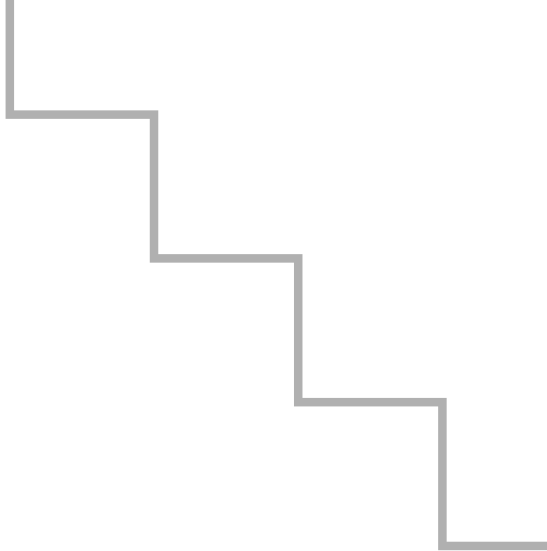




CS690

MULTI-CONDITION DIFFERENTIAL CELL-CELL COMMUNICATION INFERENCE USING SPATIAL TRANSCRIPTOMICS DATA

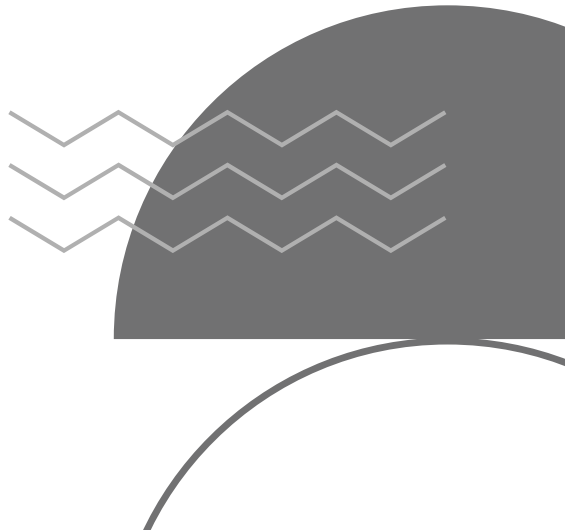
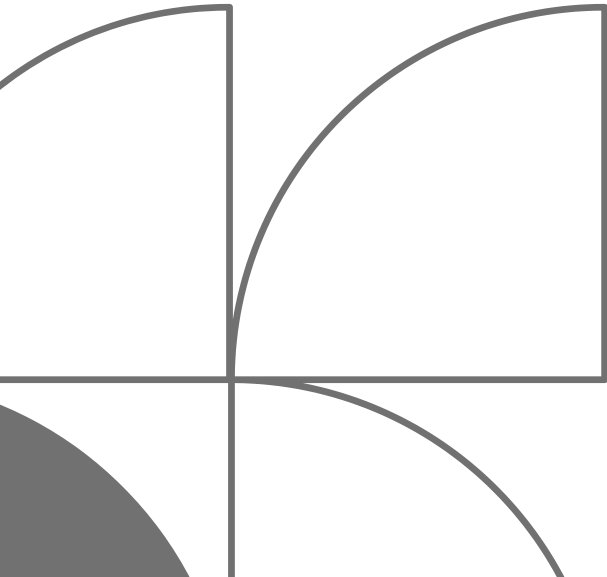
TEAM-7

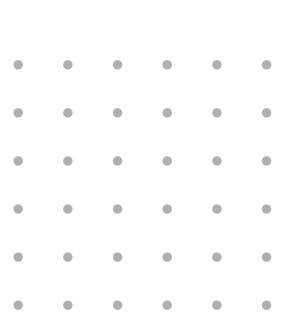
Vaneesha S Kumar

Nischay Patel

Zartaf Raza

Anshul Ranka





OBJECTIVE

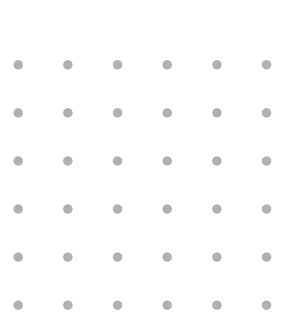
- 1) To develop a method for inferring differential communications from multiple spatial datasets
- 2) How does Cancer reshape ligand–receptor signalling programs, and what does spatial context reveal that non-spatial modelling misses?

DATA & PREPROCESSING

- INPUT Data : COMMOT LR interaction matrices
- Spot-level sender–receiver edges with coordinates
- Harmonised LR edges across conditions (only overlapping)
- Gene symbol annotation using *MyGene*

ligand	receptor	sender	receiver	score	sender_celltype	receiver_celltype	sender_x	sender_y	receiver_x	receiver_y
ENSG00000078401	ENSG00000136160	AAACTCGTC	AAACTCGTG	0.33066846	B cell	B cell	30467	12082	30467	12082
ENSG00000078401	ENSG00000136160	AAACTCGTC	ATCGCTGCC	0.00254133	B cell	B cell	30467	12082	30007	12081
ENSG00000078401	ENSG00000136160	AAAGGGCA	TTACCCTAAC	0.34488755	B cell	B cell	10481	12458	10021	12457
ENSG00000078401	ENSG00000136160	AACCATAGC	AATGTATGGC	0.35450427	B cell	B cell	7015	28842	6556	28841
ENSG00000078401	ENSG00000136160	AACGTACTC	GAGTATGCC	0.00156651	B cell	B cell	14145	22055	14374	22455
ENSG00000078401	ENSG00000136160	AACGTGCG	AACGTGCGA	0.43350026	pericyte	pericyte	21501	17267	21501	17267
ENSG00000078401	ENSG00000136160	AAGAGATGA	TAACGCTTTC	0.1237364	B cell	B cell	10709	13657	11168	13658
ENSG00000078401	ENSG00000136160	AATGGTCC	TTAGCTCTGT	0.00311034	endothelial cell of vascular tre	B cell	8180	14853	7950	15253
ENSG00000078401	ENSG00000136160	ACAAGCAG	TATCTACAGA	0.4439044	mammary gland epithelial cell	B cell	10478	14856	10018	14855
ENSG00000078401	ENSG00000136160	ACAGCGAC	ACCCATTGT	0.42656419	fibroblast	fibroblast	27233	26468	27463	26868

INPUT FORMAT



Method 1 : Mann-Whitney U Test

Non-parametric statistical test

Compares two independent distributions without assuming normality.

Evaluates whether one group tends to have larger values than the other.

When is it used?

Data are non-Gaussian, skewed, or contain outliers.

Sample sizes may be small or unequal.

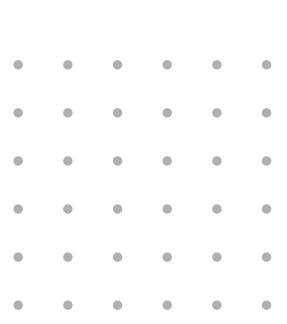
Ideal for rank-based comparison of values.

Why it fits ligand–receptor (LR) scores

LR communication scores are:

- 1.non-normal
- 2.zero-inflated
- 3.heterogeneous across spot pairs

Therefore, Mann–Whitney provides robust, assumption-free significance testing.



How did we apply it?

Objective

Quantify which LR interactions differ significantly between normal and cancer tissues.

Procedure

For each LR pair and each sender→receiver cell-type combination:

- Collect LR edge scores in cancer and normal conditions.

- Require ≥ 3 edges per condition to ensure stability.

Compute:

- Median LR score per condition

- Median-based $\log_2$ fold-change

Define significant LR interactions as:

$$|\log_2 FC| > 1, p < 0.05$$

$$\log_2 FC = \log_2 \left(\frac{\tilde{x}^{(c)} + 10^{-6}}{\tilde{x}^{(n)} + 10^{-6}} \right)$$

Results : Comparison of Differential CCC Methods

Method 1 : Mann-Whitney U Test

log2FC	pvalue	n_edges_cancer	n_edges_normal	direction	ligand_symbol	receptor_symbol
-4.770037	0.037926	222	142	Up in Normal	GRN	SORT1
1.143008	0.036589	18	20	Up in Cancer	PGF	FLT1
2.029497	0.000004	10	53	Up in Cancer	PDGFC	PDGFRA
-1.944358	0.017570	175	53	Up in Normal	IL34	CSF1R
1.250857	0.018211	8	84	Up in Cancer	ANGPTL4	CDH11
2.114535	0.035447	6	97	Up in Cancer	ANGPTL4	SDC3
-3.550658	0.000007	898	153	Up in Normal	LGALS9	CD44
-4.827635	0.038042	332	14	Up in Normal	IL16	CD4
1.166056	0.000119	147	314	Up in Cancer	GAS6	AXL
-1.565446	0.003049	211	65	Up in Normal	CSF1	CSF1R
1.342291	0.030963	15	33	Up in Cancer	PROS1	AXL
1.411631	0.013214	23	110	Up in Cancer	PDGFA	PDGFRB
1.696783	0.045465	10	70	Up in Cancer	PDGFA	PDGFRA
-5.090445	0.000248	343	105	Up in Normal	TNFSF12	TNFRSF12A

Key Differential Interactions:

- 1) Cancer tissue shows loss of immune-regulatory signalling (e.g., GRN→SORT1, IL16→CD4; strong negative log2FC)
- 2) Gain of **growth/EMT pathways**, including PGF→FLT1, PDGFC→PDGFRA, ANGPTL4→CDH11/SDC3, and GAS6/PROS1→AXL.

Differential LR network (Up in Cancer)



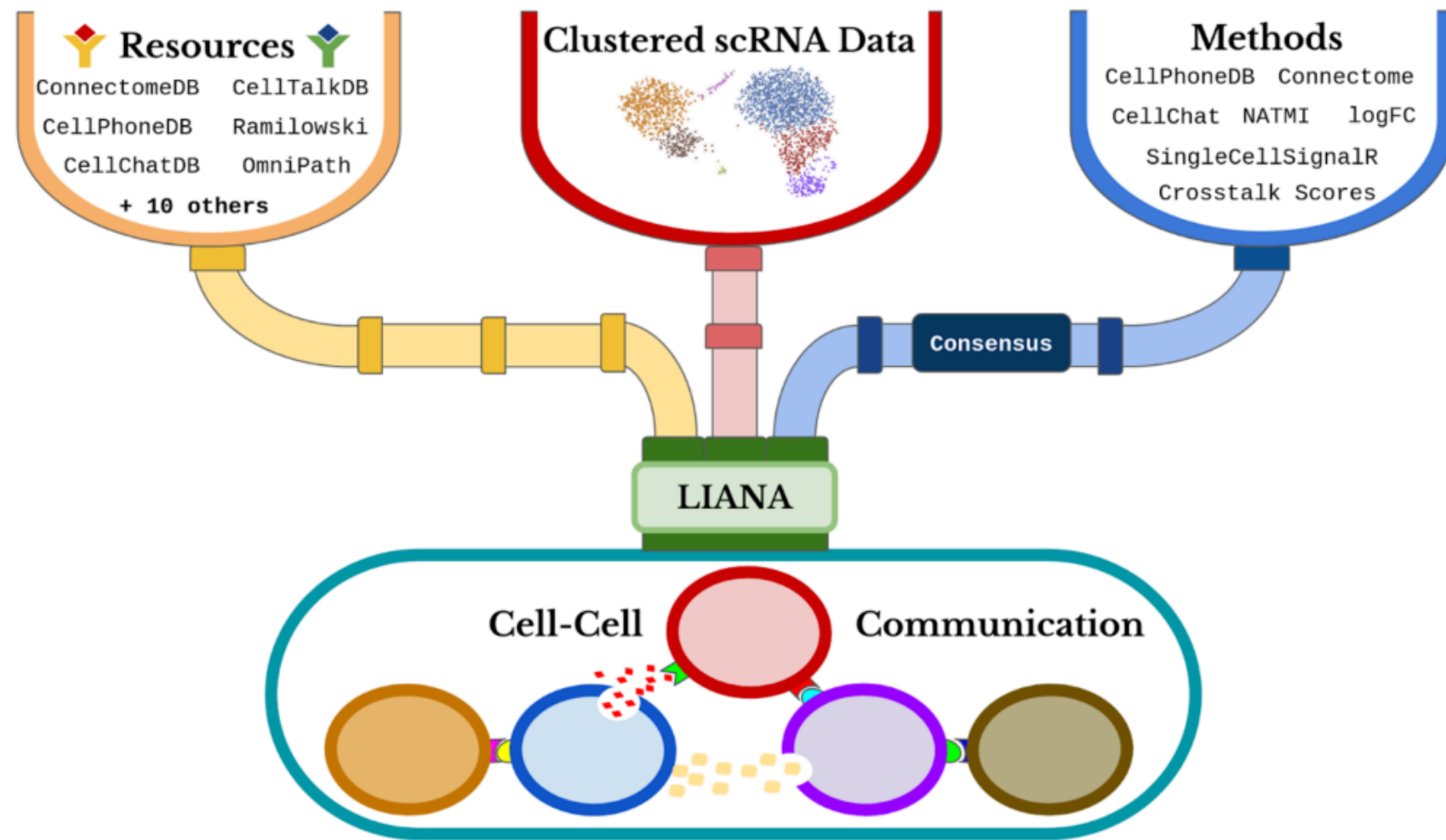
Network Modules:

The differential network resolves into three modules:

1. Angiogenic PDGF/VEGF cluster
2. AXL-centred EMT module
3. Suppressed immune module (CD4, CD44, SORT1)

Cancer is characterised by enhanced *angiogenesis* and *EMT-related signalling*, coupled with reduced *immune homeostasis*, consistent with known tumour-progression mechanisms.

How LIANA+ Infers Cell–Cell Communication



Ligand Receptor interactions are aggregated by cell type and not per spot level

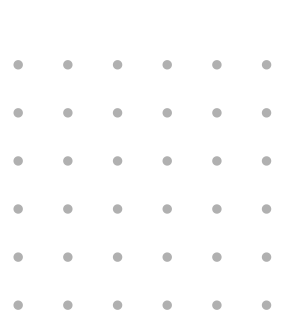
Why LIANA+ was considered

- LIANA+ provides an ensemble-based, consensus framework for ligand–receptor inference.
- Integrates multiple CCC scoring methods into a unified ranking.
- Widely used for bulk and single-cell analyses.

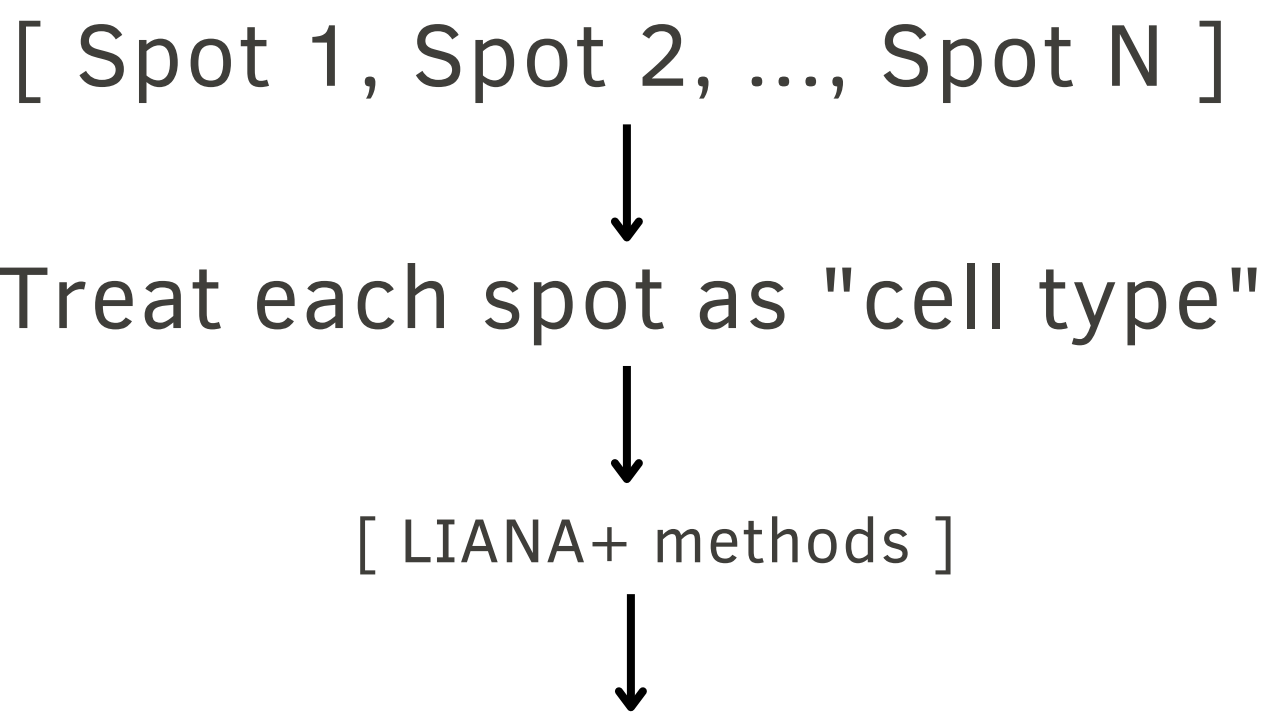
Why LIANA+ was not suitable

- LIANA+ operates on cell-type aggregated matrices, assuming each “cell group” is homogeneous.
- Attempts to bypass this violate method assumptions and produce invalid biological results.

Liana+ basically infers
Cell-type → Cell-type CCC



Why LIANA+ Cannot Perform Spot-Spot CCC in Spatial Data



The above approach if incorporated for our use case would fail because of the following reasons:

- Invalid statistics (very little group level variance)
- No incorporation of the spatial constraints
- LIANA+ does not incorporate distance constraints, neighbourhoods, or diffusion limits.

LIANA+ Assumes	What Spot-Spot Needs
Cell-type groups	Individual spatial spots
Aggregate expression	Raw spot-level expression
Population variance	Single-observation variance
Global CCC	Local, distance-aware CCC
No spatial modelling	Neighbourhood-based interactions

Method 2 : “scDisInFact” like?

Learns three latent factors:

‘Zc’ which is Shared biology

‘Zd’ which is Condition-specific effect (Unshared Biofactor)

‘Zb’ which is Batch effect (Unshared Biofactor)

Enables:

Counterfactual predictions

Condition-effect estimation

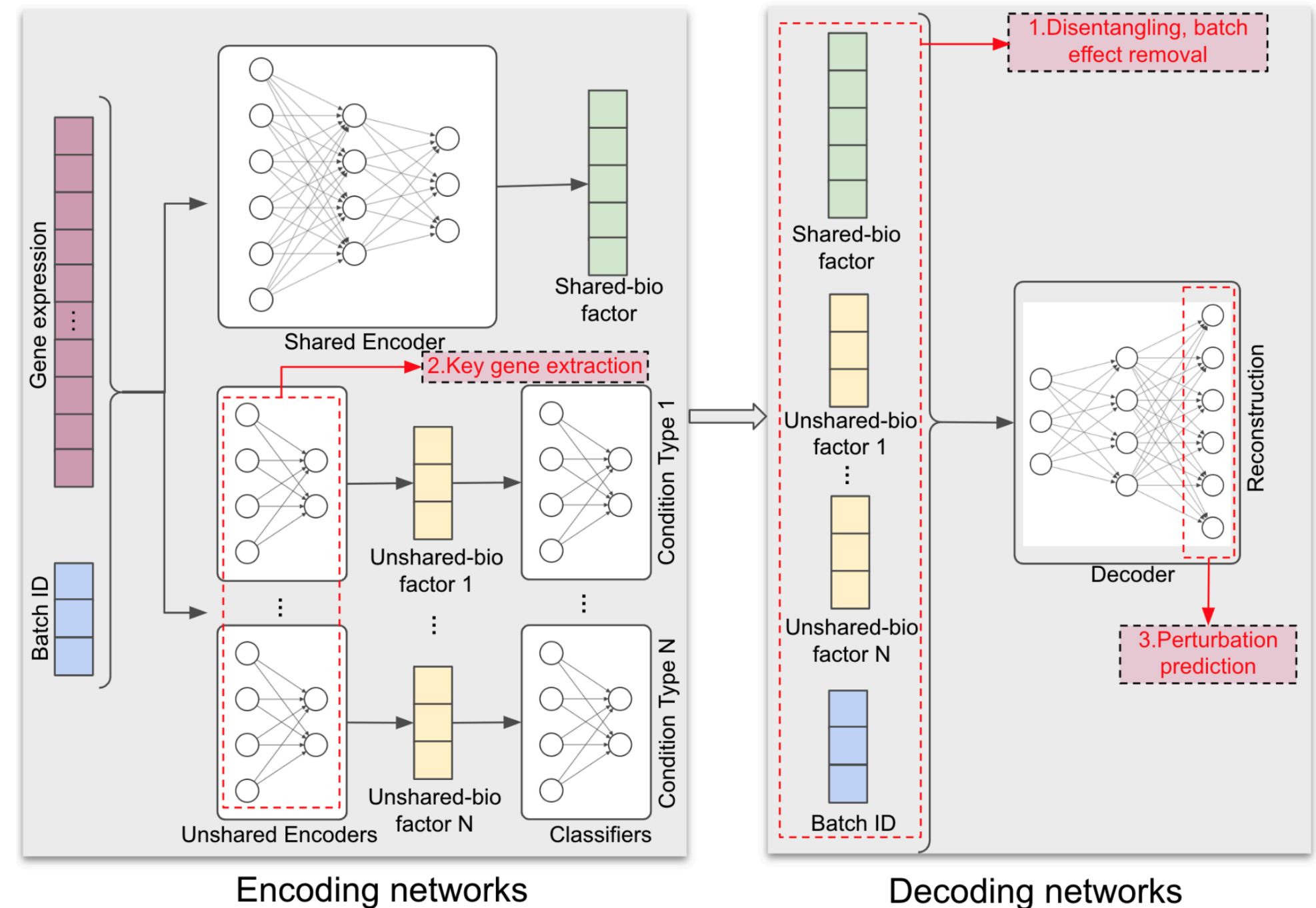
Clean separation from batch noise

Why powerful:

Separates true biology from confounders

Recovers latent gene programs

Robust differential analysis with non-linear models



A generative model that disentangles shared biology, condition effects, and batch variation from single-cell data.

Method 2A : “scDisInFact” like (only non-spatial features)

Problem

- SCDISInFact was developed for gene expression matrices (cells × genes).
- Our goal: model ligand–receptor (LR) communication, not gene counts.
- Communication is defined at the sender–receiver edge level.

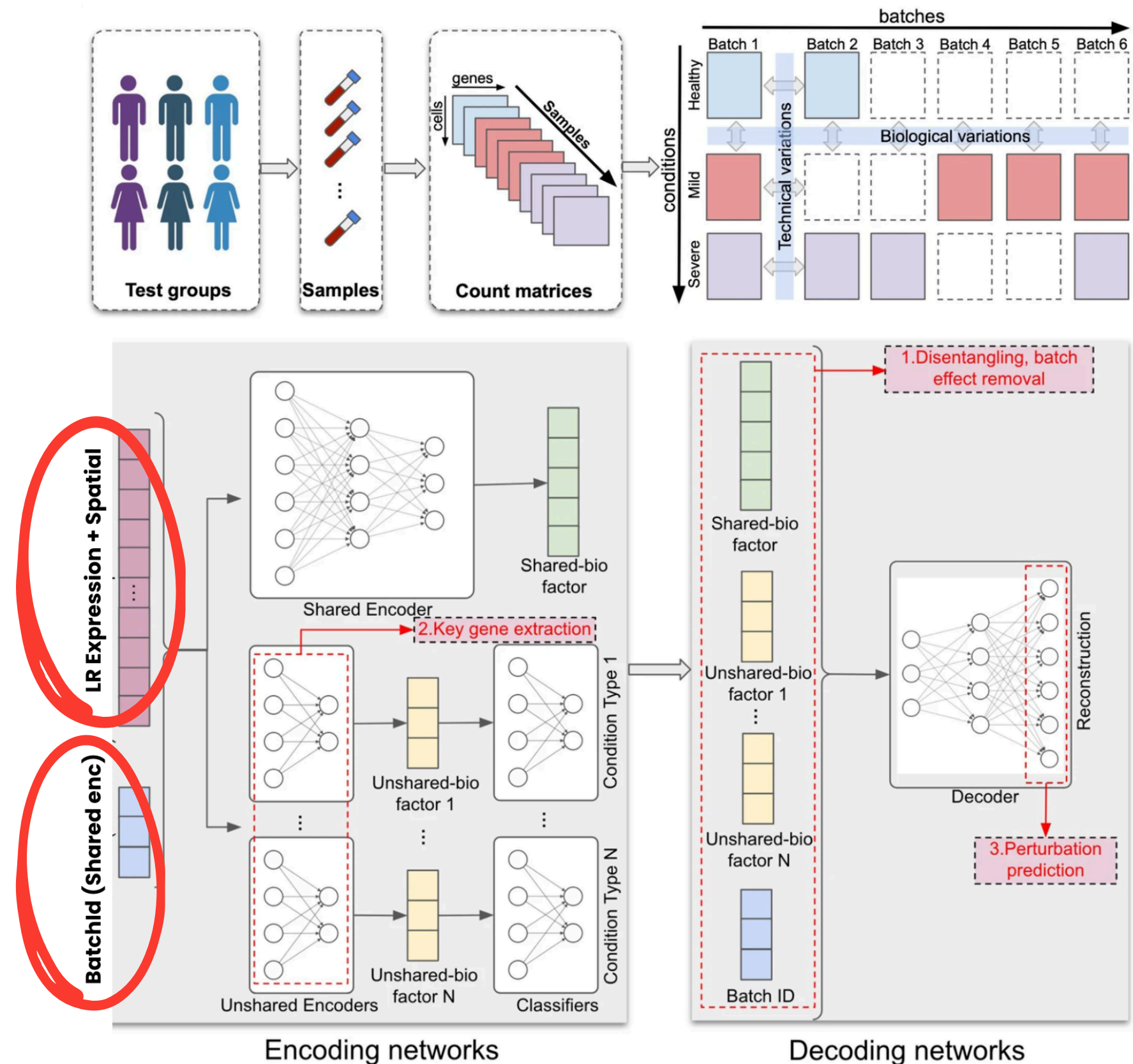
Our Adjustment

- Replace original input matrix:
cells × genes with **sender_receiver × LR_pairs**
- Each row becomes a biological interaction event:
 - Sender cell to Receiver cell
 - Vector of Ligand-Receptor (LR) scores for all LR pairs

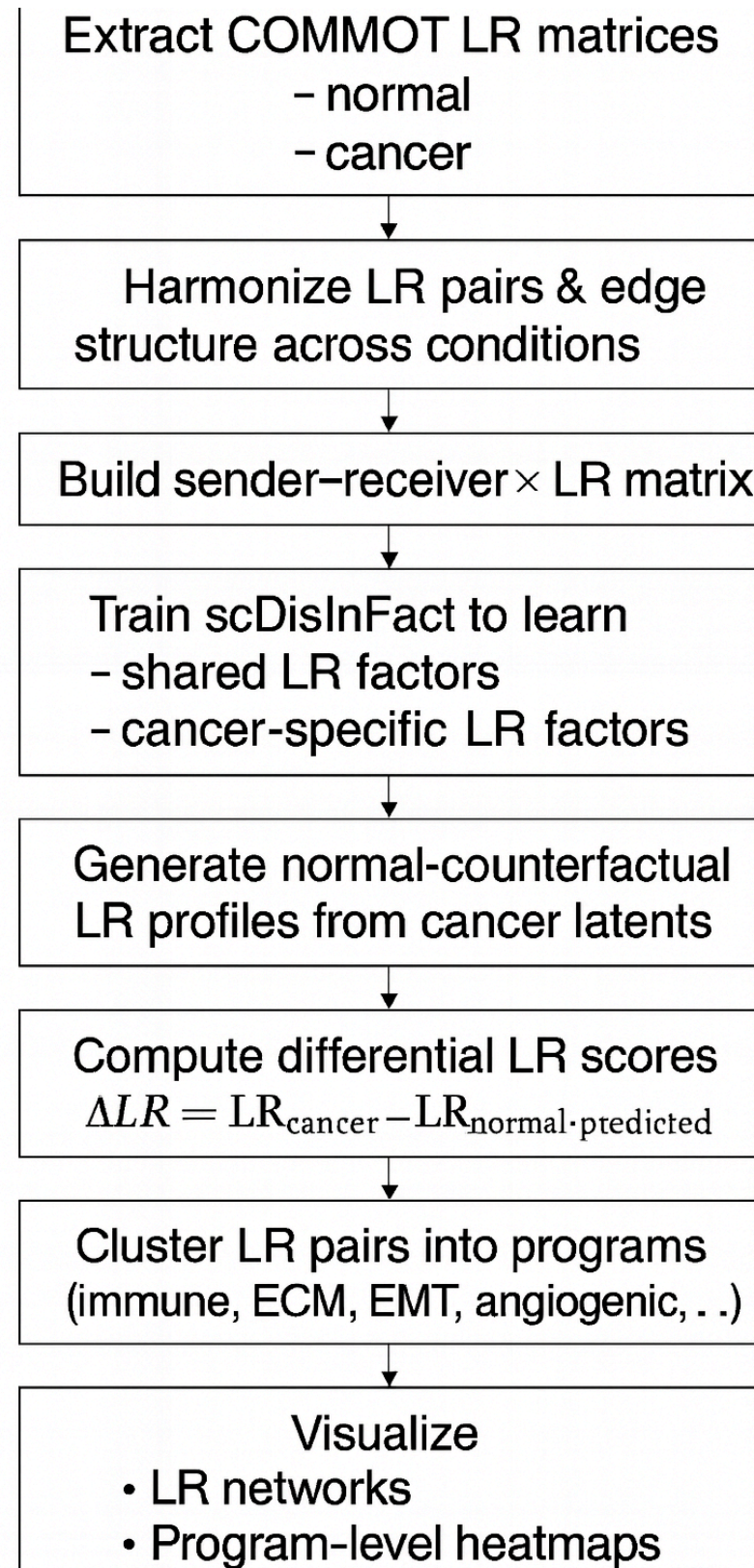
Since, SCDISInFact is modality-agnostic, it only requires:

- A feature matrix,
- Condition labels, and
- A continuous input domain.

Our Input : LR edge matrices satisfy these assumptions.



Analysis Pipeline

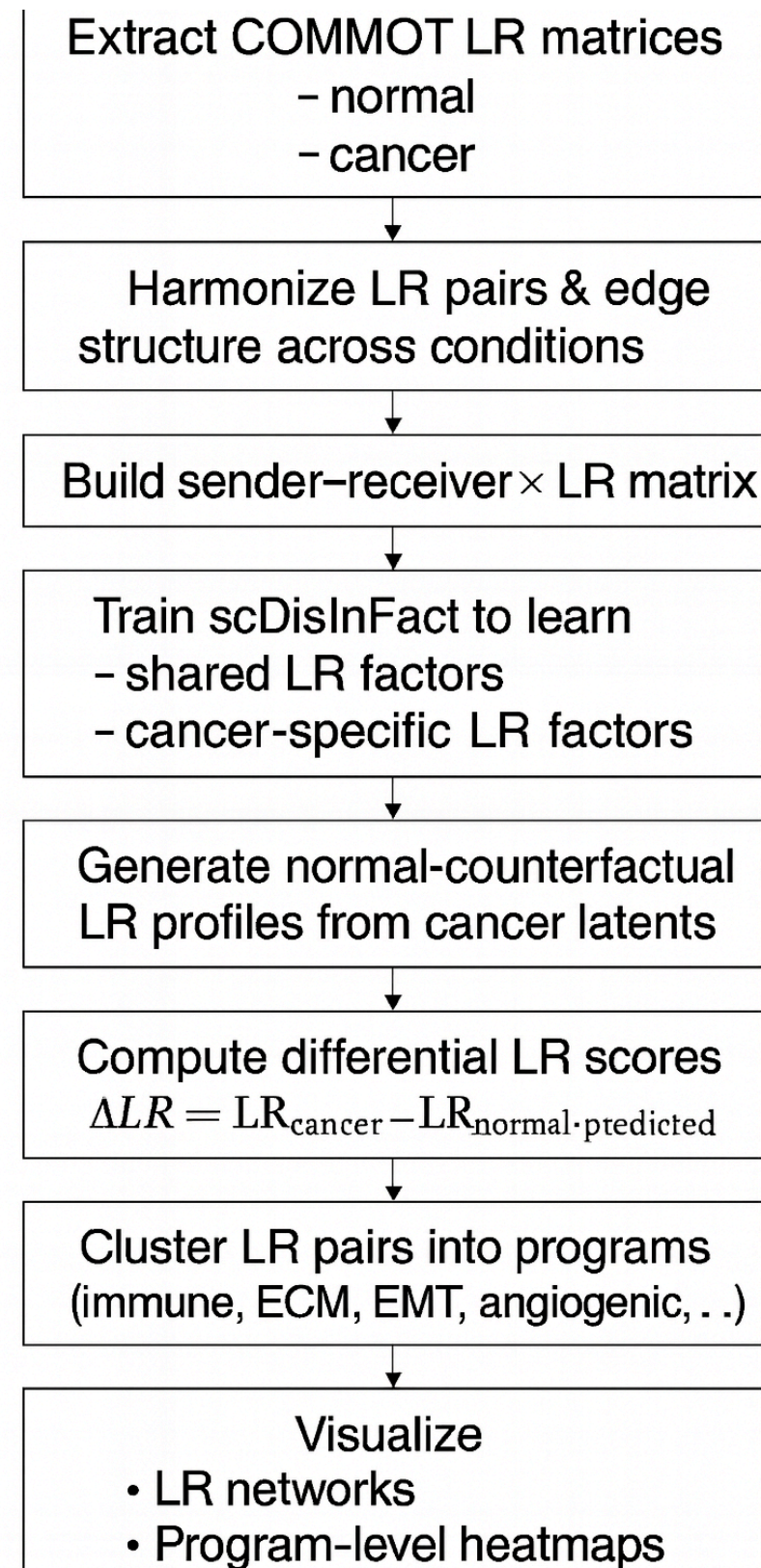


Differential LR scores: $\Delta LR = LR_{\text{cancer}} - LR_{\text{normal-predicted}}$

- 1) **Stromal/ECM** (Tumors Remodel Their Surroundings)
- 2) **Growth/Survival** (Tumor Cells Activate Pro-Proliferative Signaling)
- 3) **EMT/Invasion** (Tumors Acquire Migratory Characteristics)
- 4) **Immune** (Tumors Rewire Immune Microenvironment)
- 5) **Angiogenic** (Tumors Require New Blood Vessels)

This Pipeline is used to incorporate both **Spatial** and **Non Spatial** features

Method 2B : “scDisInFact” like (spatial+non spatial features)



+

EUCLIDEAN DISTANCE

Raw sender→receiver distance

Models physical feasibility of ligand diffusion and receptor access

Log-Distance

Raw sender→receiver distance

Captures non-linear decay in ligand influence with spatial separation

Neighbourhood Hotspot Scores

Compute k-nearest neighbours around each spot (sender & receiver).

Aggregate LR activity within neighbourhoods to obtain:

Sender hotspot score
Receiver hotspot score

Combined Edge-Level Spatial Score

Average of sender + receiver hotspot scores

Represents local communication density

Importance Scores and Counter-Factual Prediction

Without Spatial

```
array([0.03890562, 0.03129333, 0.13546471, 0.04069709, 0.03435564,
       0.03347379, 0.03858469, 0.03213444, 0.03554172, 0.0309734 ,
       0.033644 , 0.08844993, 0.06138325, 0.05715941, 0.04118221,
       0.05261873, 0.05410535, 0.4605851 , 0.0701035 , 0.08650023,
       0.07454513, 0.04895443, 0.02696123, 0.03506033, 0.03795723,
       0.03301947, 0.0420945 , 0.03361221, 0.03041674, 0.06340052,
       0.03179644, 0.03789425, 0.0305996 , 0.03552901, 0.03875551,
       0.02881623, 0.0366394 , 0.08811662, 0.07267613, 0.05585627,
       0.03589614, 0.04031137, 0.03179843, 0.05200454, 0.03451099,
       0.2984919 , 0.06061924, 0.02769059, 0.05442096, 0.06752266,
       0.02748072, 0.0380659 , 0.03639613, 0.12912144], dtype=float32)
```

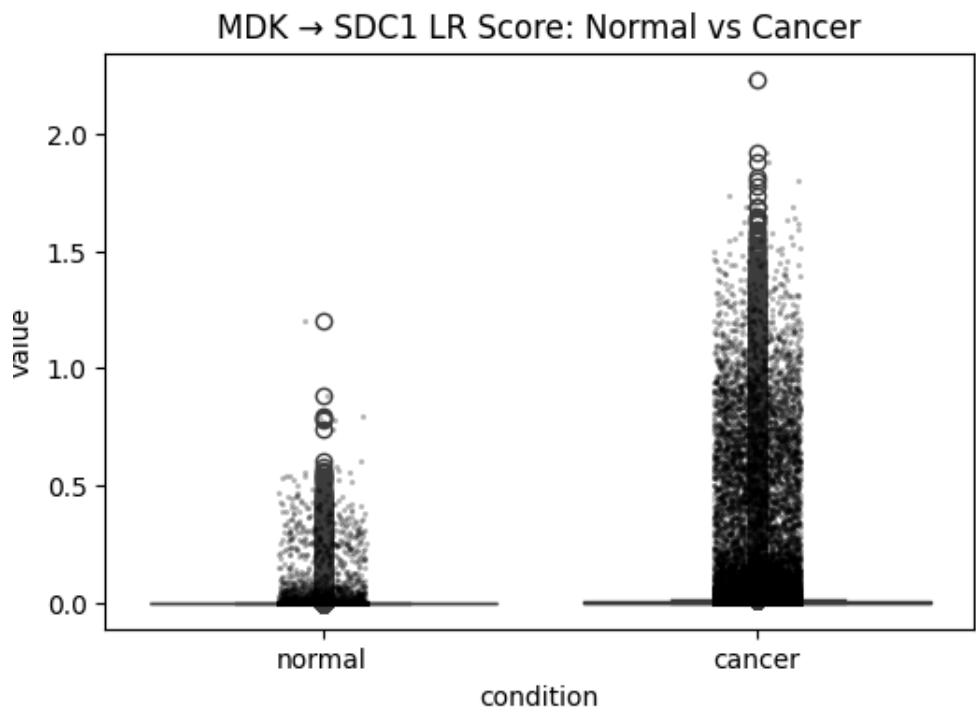
0	ligand_symbol	receptor_symbol	LR_symbolic
0.460585	MDK	SDC1	MDK → SDC1
0.298492	LGALS9	CD44	LGALS9 → CD44
0.135465	GRN	SORT1	GRN → SORT1
0.129121	TNFSF12	TNFRSF12A	TNFSF12 → TNFRSF12A

With Spatial

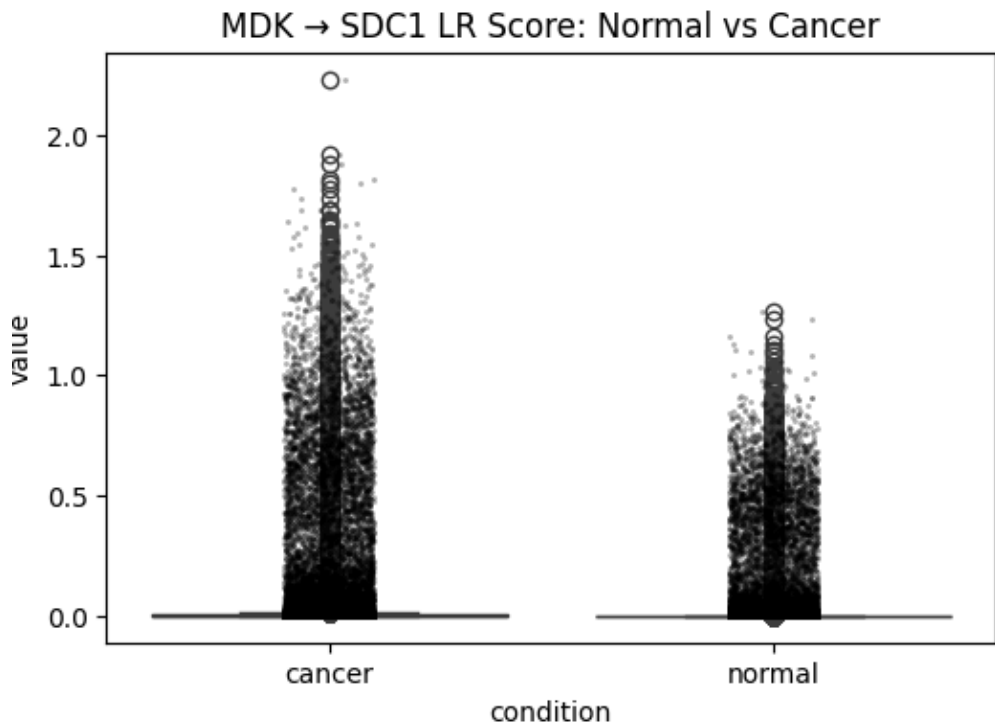
```
array([0.02860452, 0.0347073 , 0.22224857, 0.04950456, 0.02512795,
       0.03890773, 0.03987482, 0.0301917 , 0.04151778, 0.04636892,
       0.03614116, 0.10303199, 0.08682573, 0.05205829, 0.05081248,
       0.05365146, 0.0689052 , 0.57304543, 0.07656815, 0.09795426,
       0.07842883, 0.05164757, 0.02320117, 0.04158655, 0.02670755,
       0.03551676, 0.03474435, 0.03265746, 0.04194003, 0.06713592,
       0.03713068, 0.04575158, 0.03431357, 0.03347751, 0.03394548,
       0.03369703, 0.03067617, 0.08911544, 0.05837534, 0.05939914,
       0.04316936, 0.0414051 , 0.03952237, 0.04238449, 0.04175785,
       0.3934055 , 0.07944796, 0.03489893, 0.06168114, 0.07535619,
       0.03171017, 0.03486326, 0.03762843, 0.15225849, 0.49450696],
      dtype=float32)
```

0	ligand_symbol	receptor_symbol	LR_symbolic
0.573045	MDK	SDC1	MDK → SDC1
0.393405	LGALS9	CD44	LGALS9 → CD44
0.222249	GRN	SORT1	GRN → SORT1
0.152258	TNFSF12	TNFRSF12A	TNFSF12 → TNFRSF12A

Original Dataset



With CounterFactual



Results : Comparison of Differential CCC Methods

(Method 2A) VS (Method 2B)

Rank	Ligand	Receptor	LR Pair
1	LGALS9	CD44	LGALS9 → CD44
2	IL34	CSF1R	IL34 → CSF1R
3	NAMPT	ITGA5+ITGB1	NAMPT → ITGA5+ITGB1
4	IL16	CD4	IL16 → CD4
5	GRN	SORT1	GRN → SORT1
6	MDK	SDC1	MDK → SDC1
7	NAMPT	INSR	NAMPT → INSR
8	CXCL12	CXCR4	CXCL12 → CXCR4
9	MDK	LRP1	MDK → LRP1
10	MDK	SDC2	MDK → SDC2

Without Spatial

Rank	Ligand	Receptor	LR Pair
1	NAMPT	ITGA5+ITGB1	NAMPT → ITGA5+ITGB1
2	MDK	SDC1	MDK → SDC1
3	MDK	LRP1	MDK → LRP1
4	GRN	SORT1	GRN → SORT1
5	IL16	CD4	IL16 → CD4
6	LGALS9	CD44	LGALS9 → CD44
7	MDK	SDC2	MDK → SDC2
8	VEGFB	FLT1	VEGFB → FLT1
9	TNFSF12	TNFRSF12A	TNFSF12 → TNFRSF12A
10	CXCL12	CXCR4	CXCL12 → CXCR4

With Spatial

Non-Spatial Model Captures

Global cancer-associated signalling
(LGALS9 → CD44, IL34 → CSF1R,
GRN → SORT1)

Broad immunosuppressive programs
(LGALS9 → CD44, IL16 → CD4)

Diffuse chemokine axes
(CXCL12 → CXCR4)

Shared Across Both Models

MDK-driven ECM remodelling
(MDK → SDC1, MDK → SDC2, MDK → LRP1)

Integrin-mediated cell-matrix adhesion
(NAMPT → ITGA5+ITGB1)

GRN/SORT1 survival signalling
(GRN → SORT1)

CXCL12-driven migration
(CXCL12 → CXCR4)

Spatial Model Captures

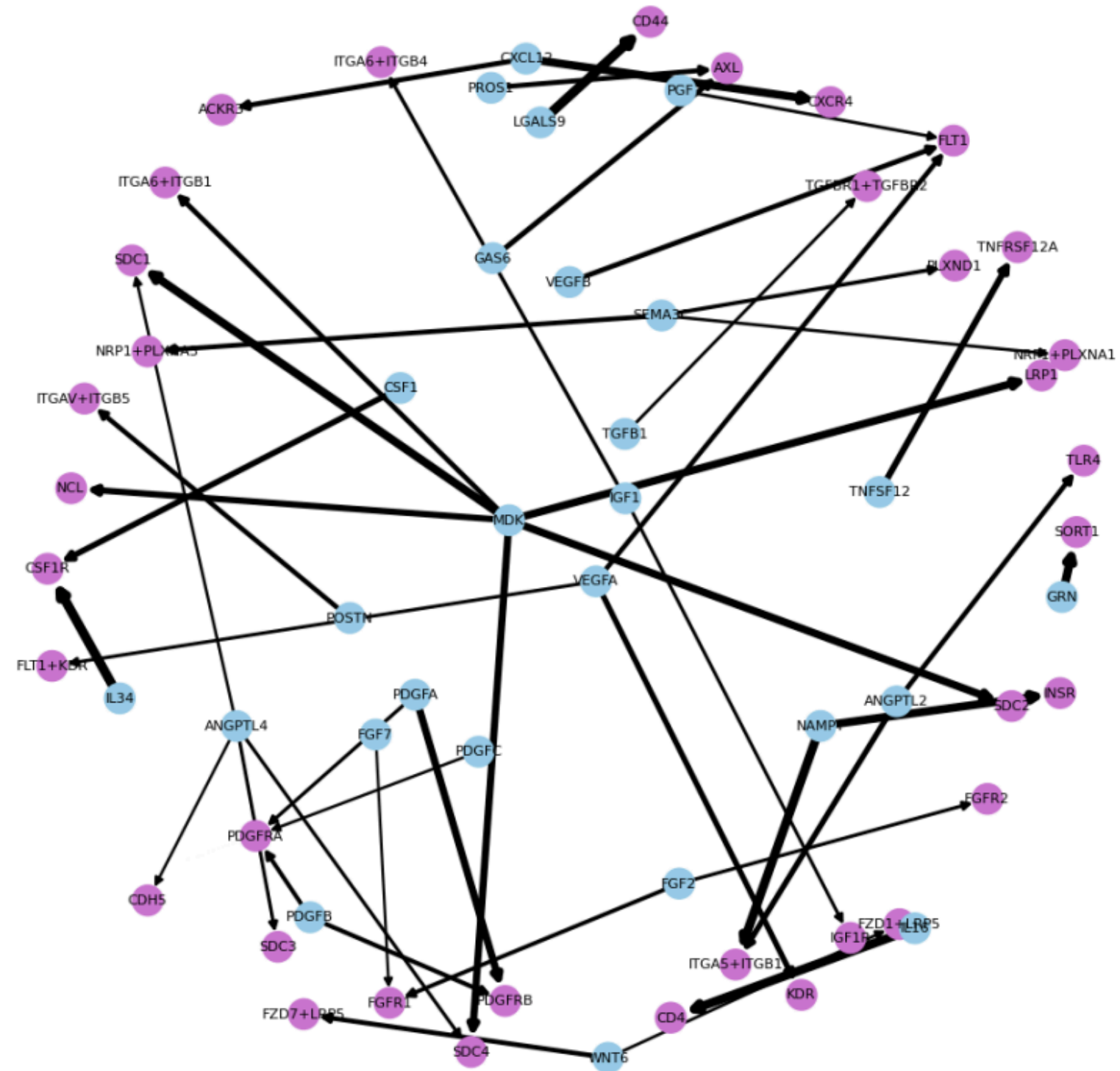
Localised adhesion and ECM remodelling
(NAMPT → ITGA5+ITGB1, MDK → SDC1,
MDK → SDC2)

Spatially constrained angiogenic signals
(VEGFB → FLT1, MDK → LRP1)

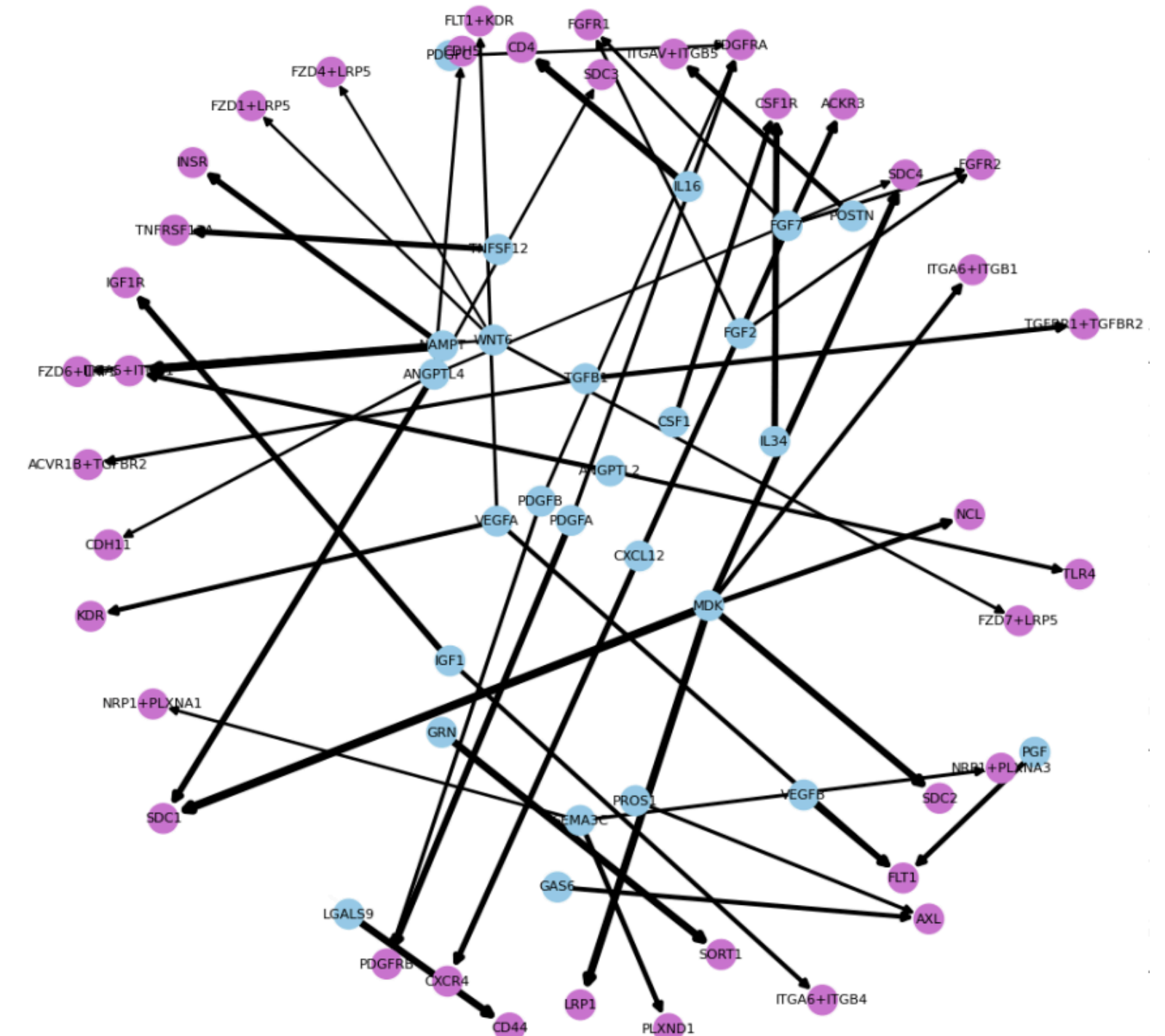
Contact-dependent inflammatory pathways
(TNFSF12 → TNFRSF12A)

High-resolution TME wiring
(GRN → SORT1, MDK → LRP1, CXCL12 →
CXCR4)

A 6x6 grid of 36 dots, arranged in 6 rows and 6 columns. The dots are small, solid black circles. The grid is centered on the page.

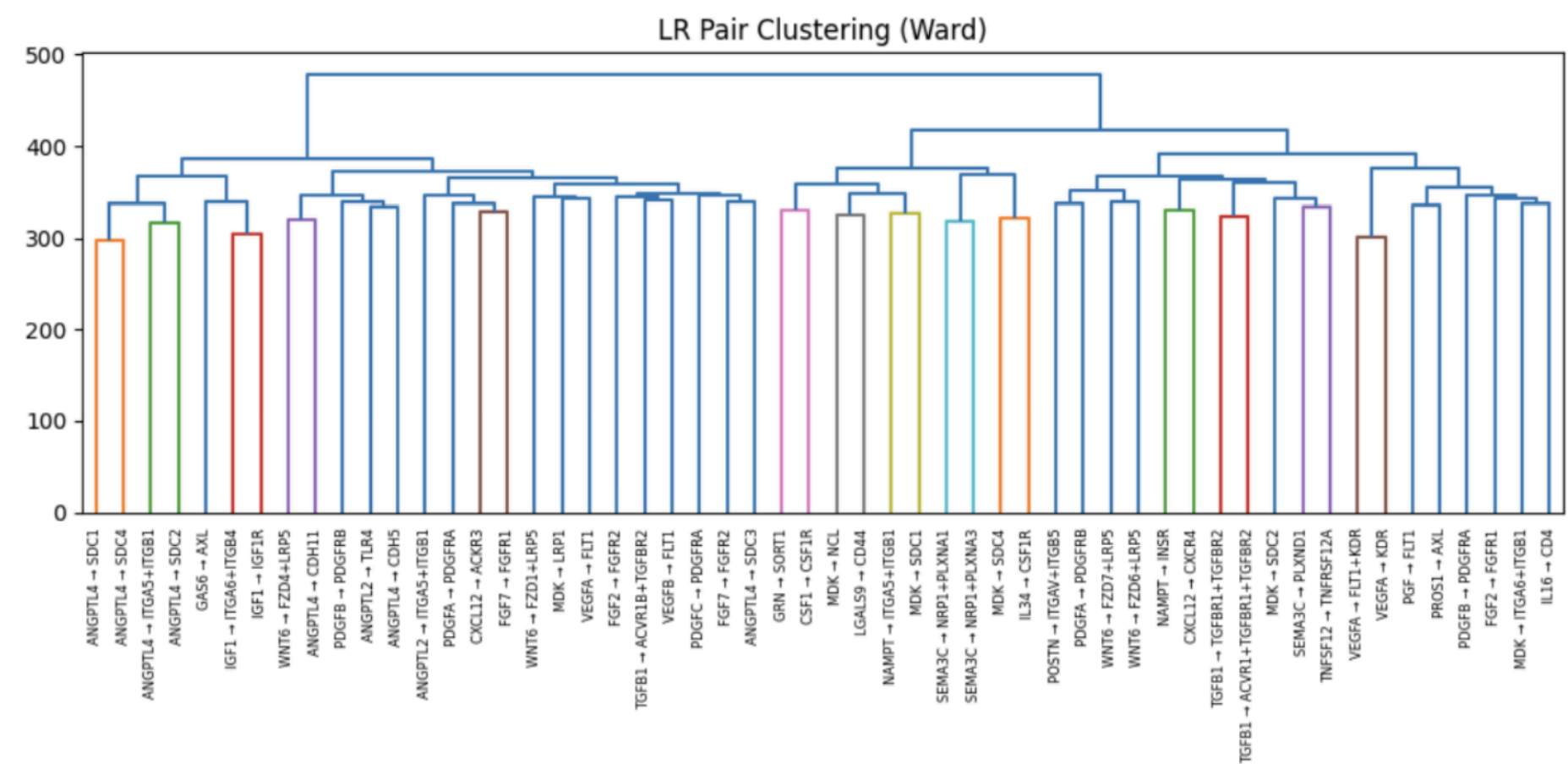


Without Spatial



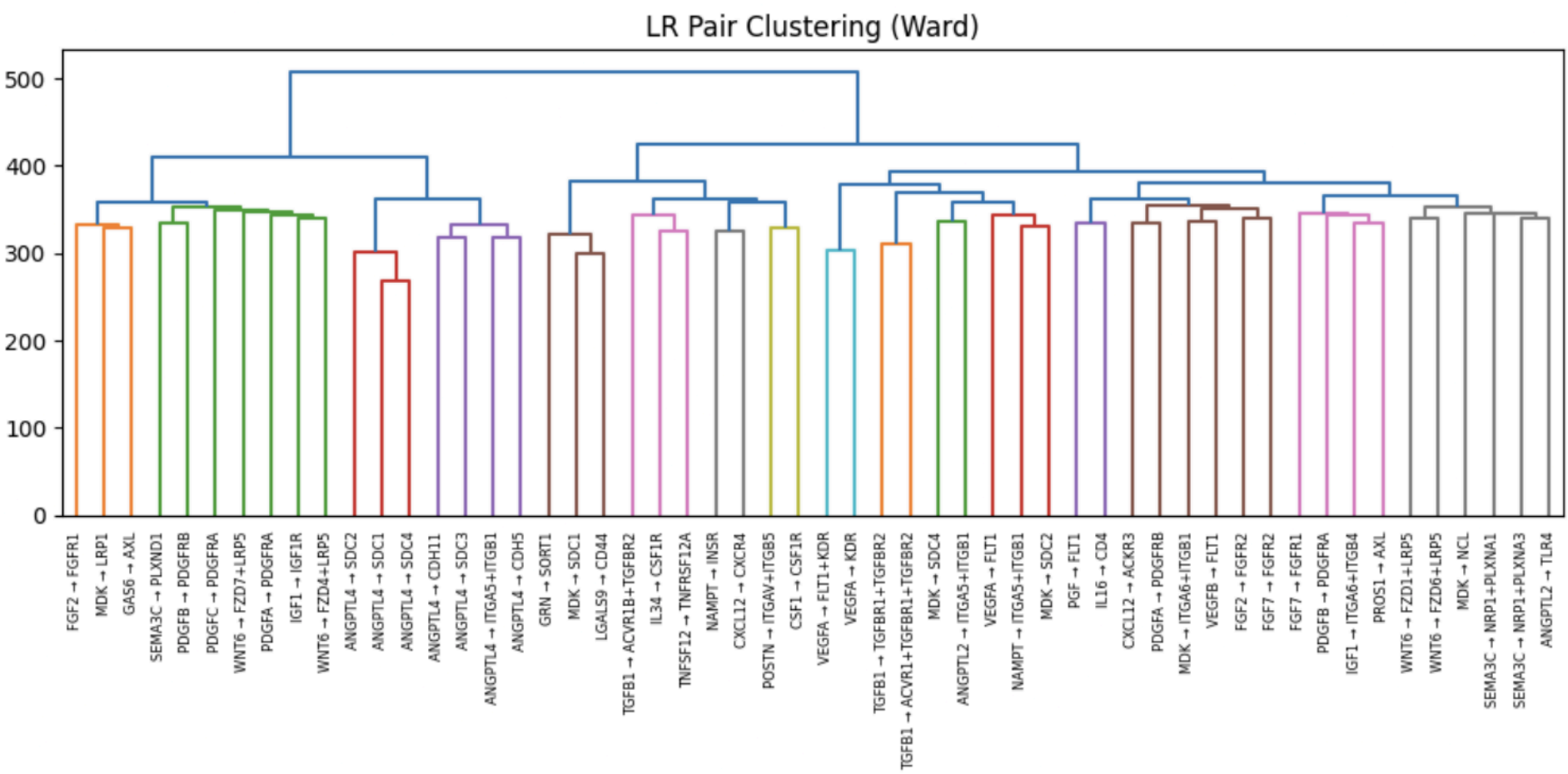
With Spatial

Results : Comparison of Differential CCC Methods (Method 2A) VS (Method 2B)



Without Spatial

Hierarchical clustering grouped LR pairs into distinct biological modules: *a)EMT/invasion, b)stromal–ECM, c)immune regulation, and d)growth-factor signalling*; indicating that the ligand–receptor landscape organises into coherent mechanistic “programs.”



With Spatial

Spatially informed SCDIsInFact revealed sharper, locally organised LR structure. Clustering of spatial LR pairs produced clear EMT/invasion, stromal–ECM, immune, and angiogenic modules, with stronger boundaries than the non-spatial model showing that physical neighbourhood constraints shape co-occurring signalling in the tumour microenvironment.

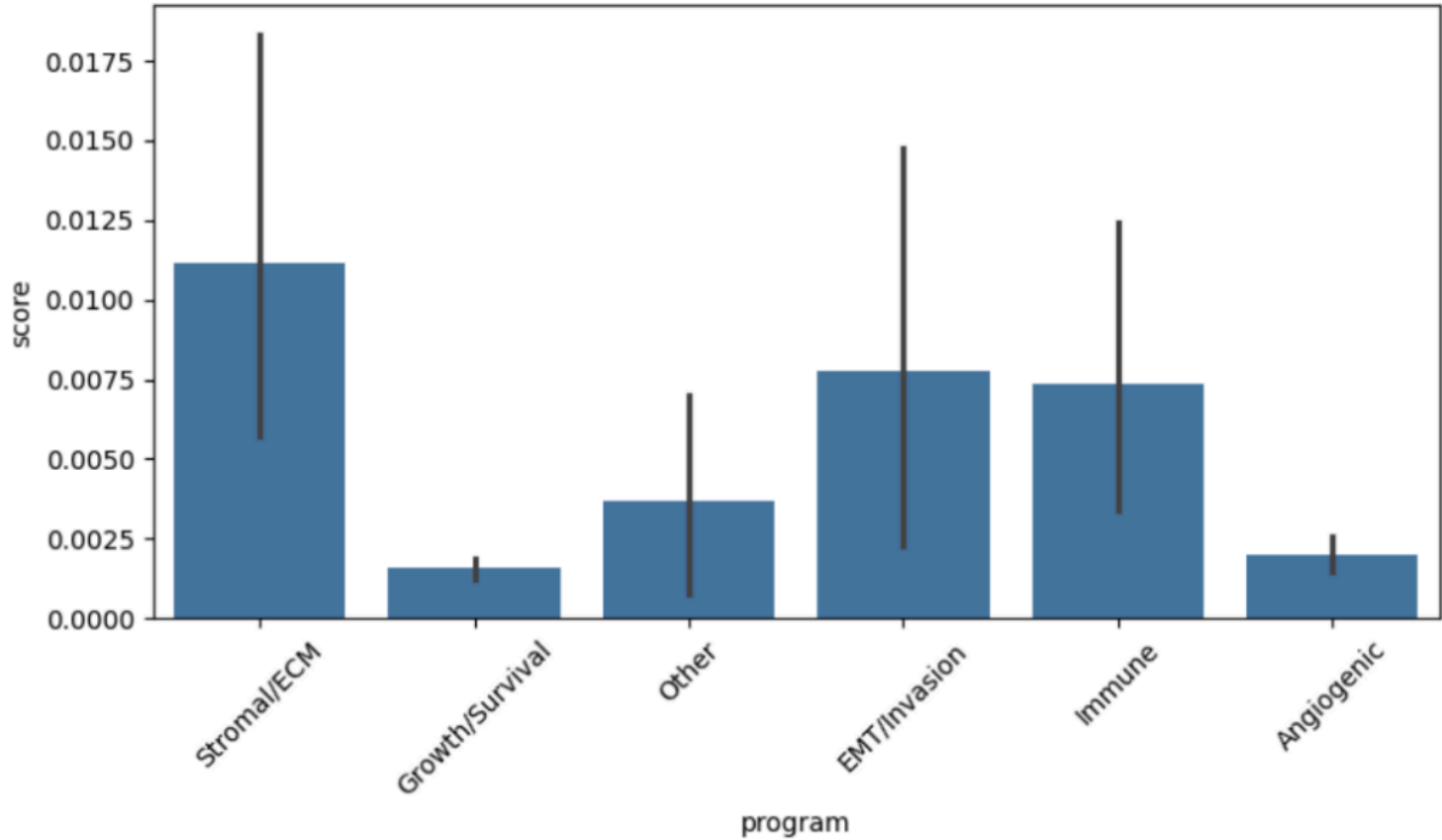
The two dendrograms show different LR-pair similarity structures with shifts in cluster height, and grouping (tighter in spatially informed) which indicates possible changes in the underlying LR activity patterns between the two conditions in both the methods.

Results : Comparison of Differential CCC Methods

(Method 2A) VS (Method 2B)

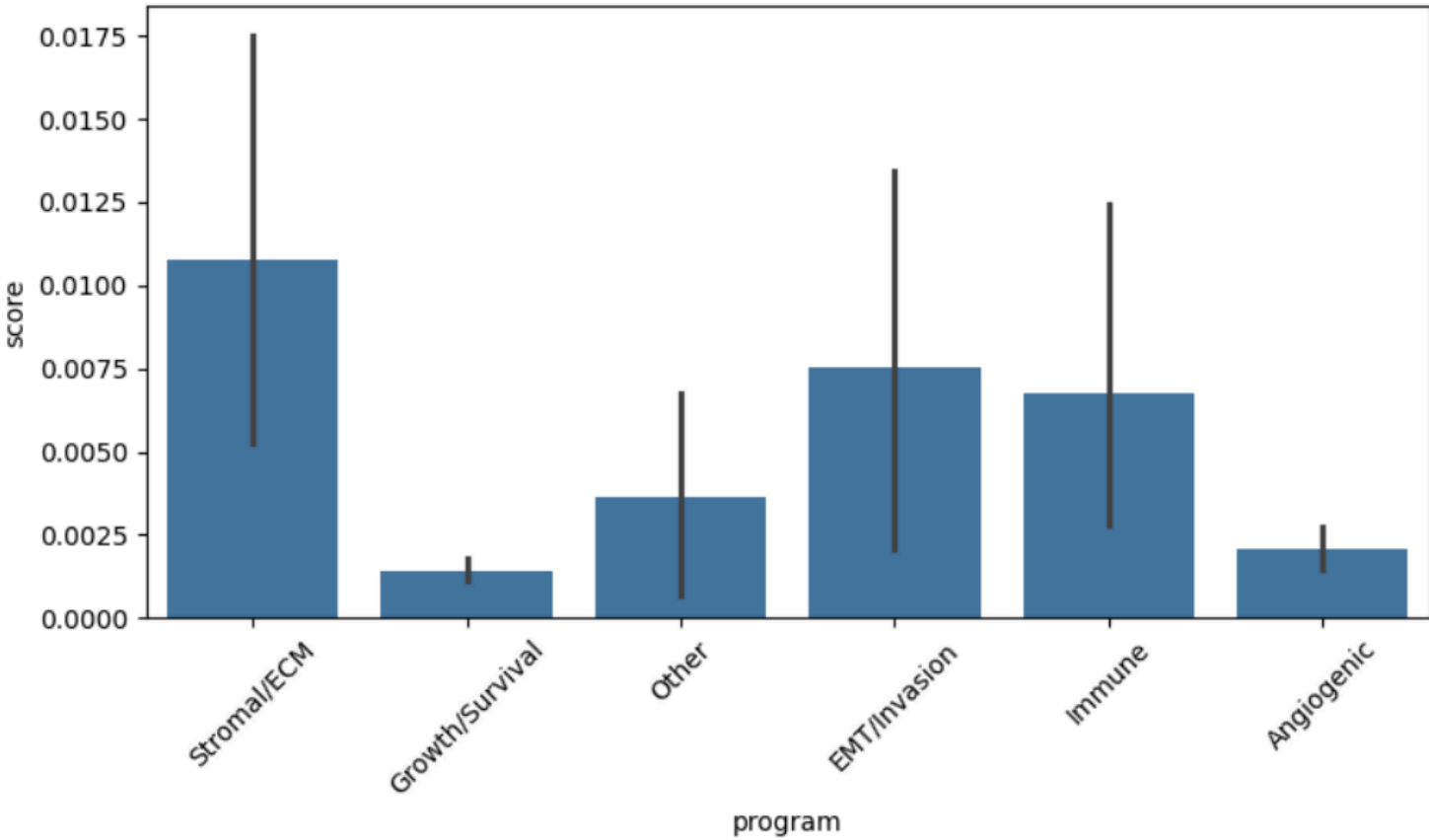
Without Spatial

Average LR Activity per Program



With Spatial

Average LR Activity per Program



Stromal/ECM Program

Highest mean activity in both plots, confirming strong cancer-associated ECM remodelling.

Wide variance indicates that ECM signalling varies substantially across edges.

Dominant LR pairs: MDK→SDC1/SDC2/SDC4, ANGPTL4→SDC3, IGF1→integrins.

EMT/Invasion Program

The EMT/Invasion program shows comparable mean activation.

This consistency reflects the fact that EMT signalling—driven by pathways such as SEMA3C–PLXND1, TGFB1–TGFB1R, and ANGPTL4–CDH11—is a global property of malignant tissue

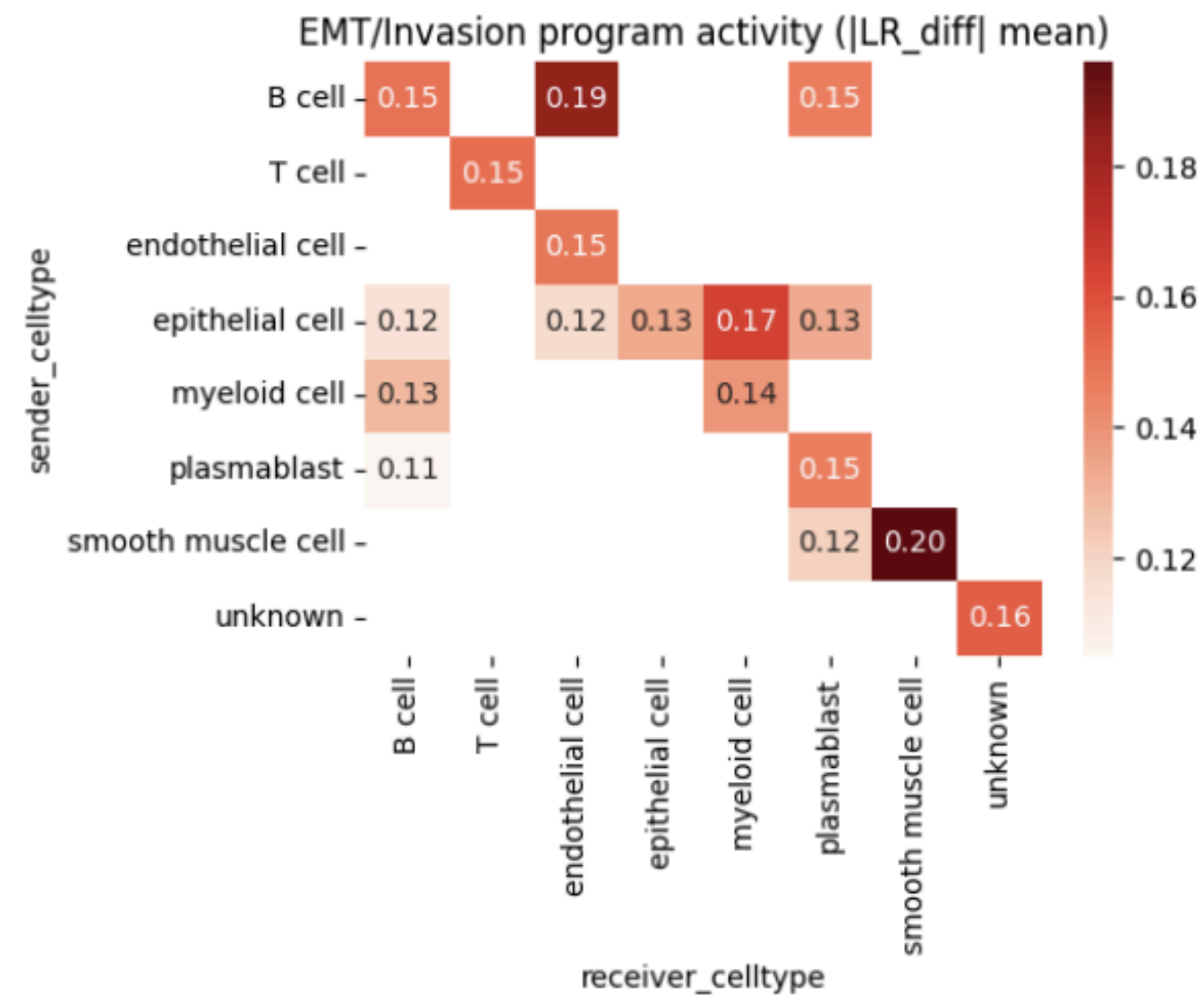
Immune Program

The Immune program also maintains a similar overall magnitude in both models.

This agreement suggests that immune dysregulation in cancer is widespread and not restricted to specific spatial niches.

Results : Comparison of Differential CCC Methods

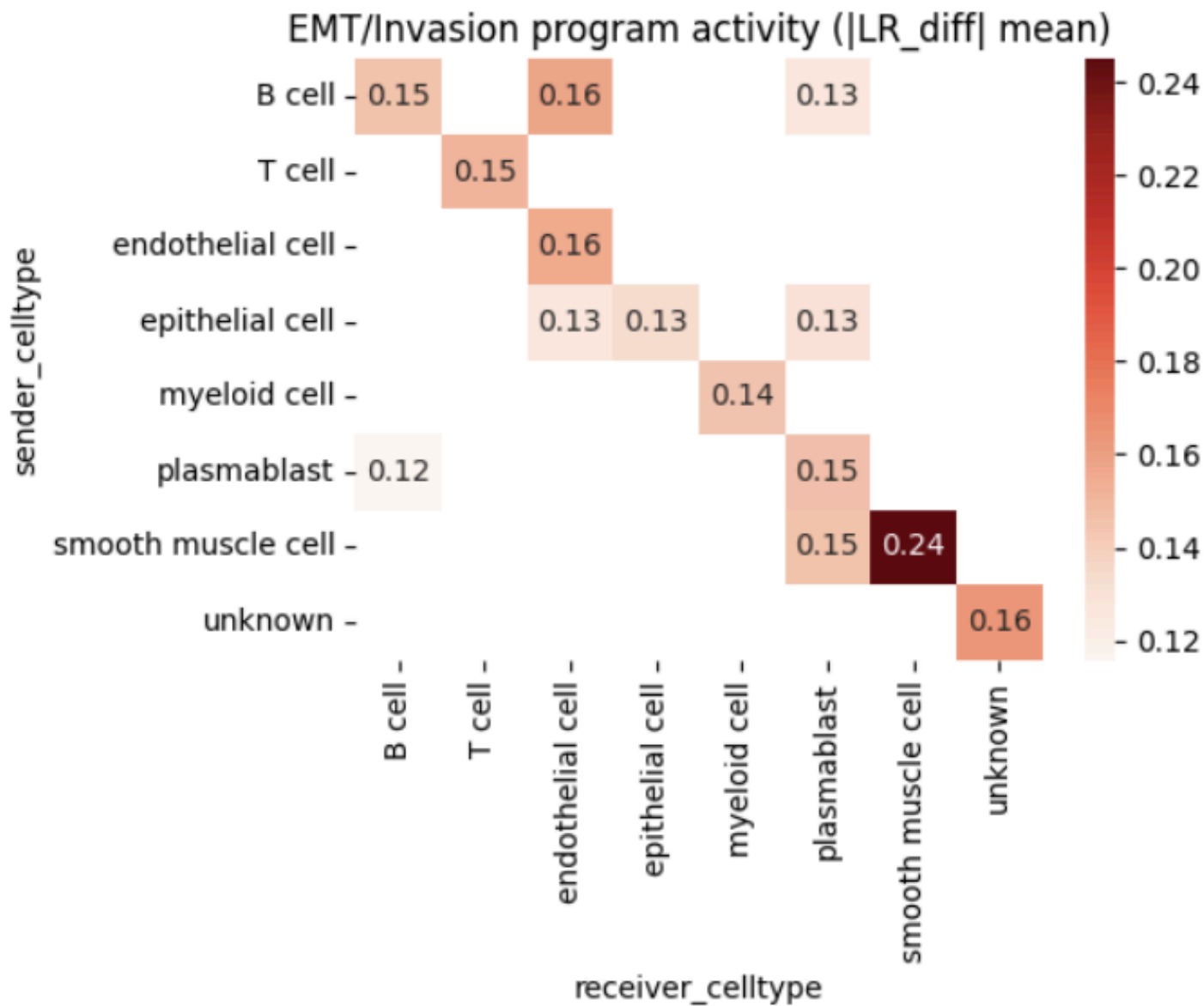
(Method 2A) VS (Method 2B)



Without Spatial

Observation: Epithelial -> Myeloid interaction (Score 0.17) disappears in the Spatial model.

Inference: High expression abundance != interaction. Spatial segregation (compartmentalization) prevents active signaling in this sample.



With Spatial

Observation: Smooth Muscle autocrine loops intensify (0.20 -> 0.24).

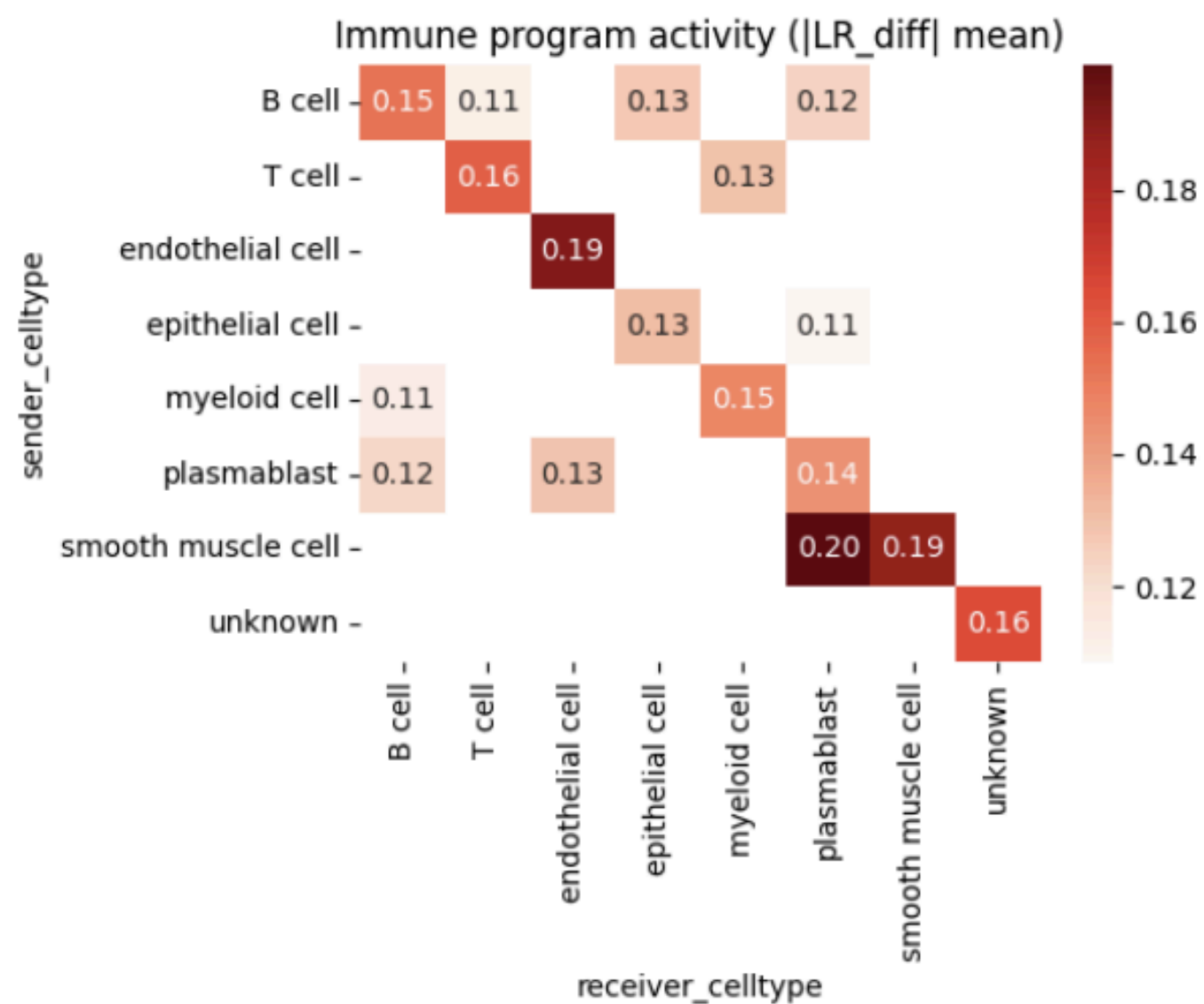
Inference: EMT and remodeling signals are driven by dense, structural niches (vessels/stroma) rather than diffuse tissue-wide broadcasting.

Observation: B-cell broad broadcasting is reduced; localized signaling remains.

Inference: Suggests restriction to Tertiary Lymphoid Structures (TLS) rather than diffuse infiltration.

Results : Comparison of Differential CCC Methods

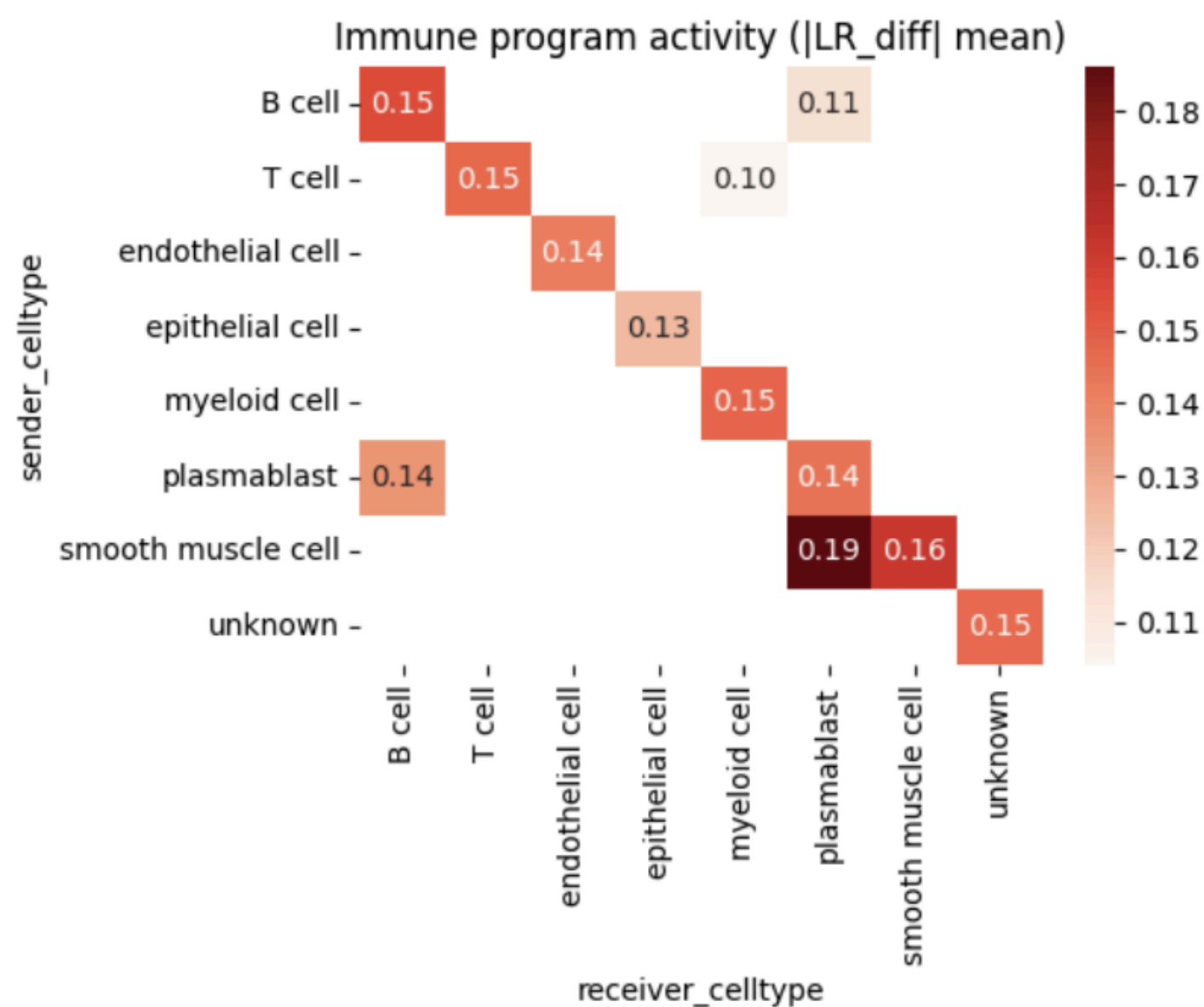
(Method 2A) VS (Method 2B)



Without Spatial

Observation: B-cell → Epithelial crosstalk is removed in the Spatial model.

Bio-Inference: B-cells are physically segregated in Tertiary Lymphoid Structures (TLS) and do not directly signal to tumor islands, correcting a non-spatial overexpression artifact.



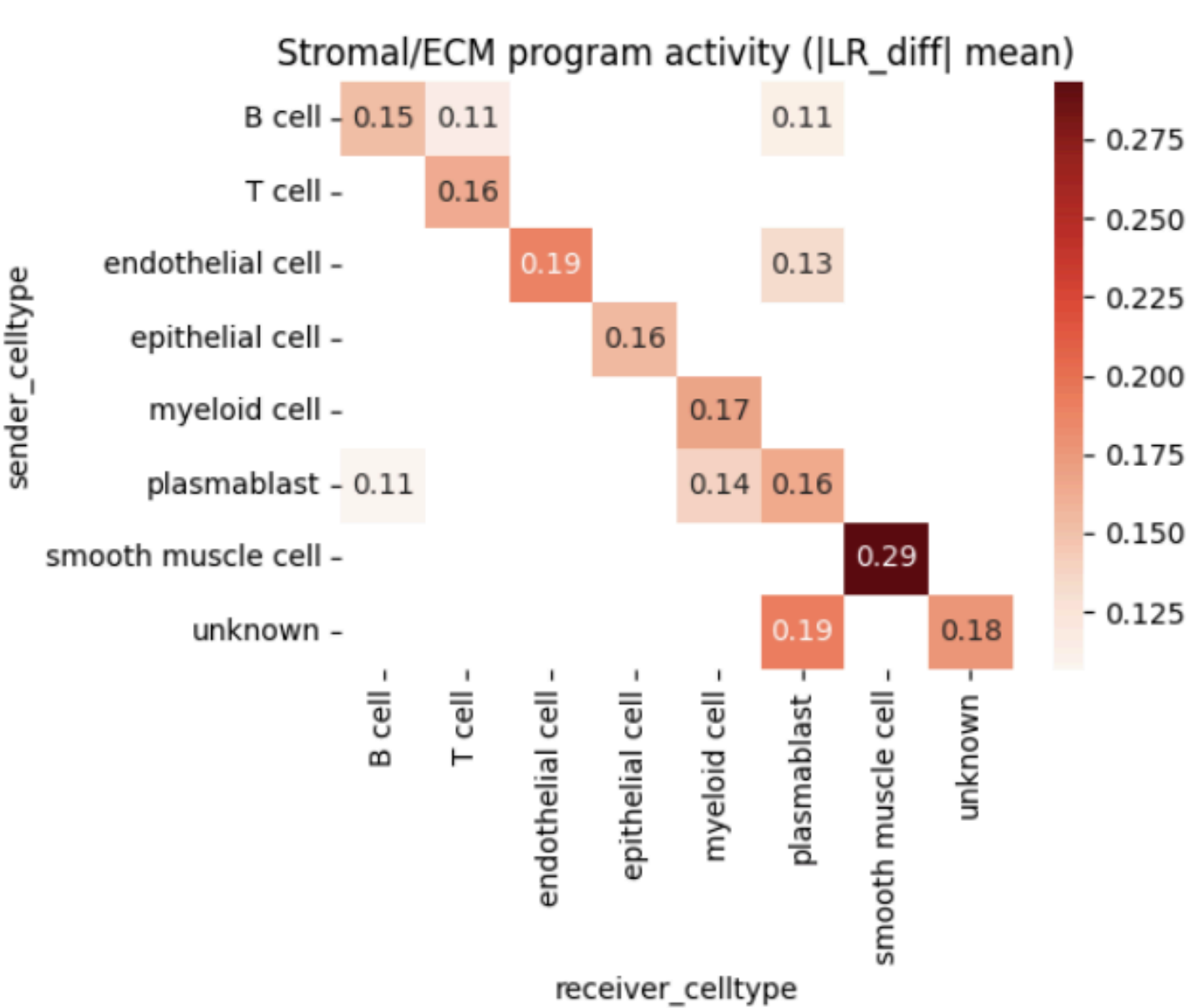
With Spatial

Observation: Endothelial autocrine immune scores drop (0.19 → 0.14).

Bio-Inference: Suggests tumor vessels may be "anergic" (immunologically quiet) locally, despite high global expression, hindering effective immune infiltration.

Results : Comparison of Differential CCC Methods

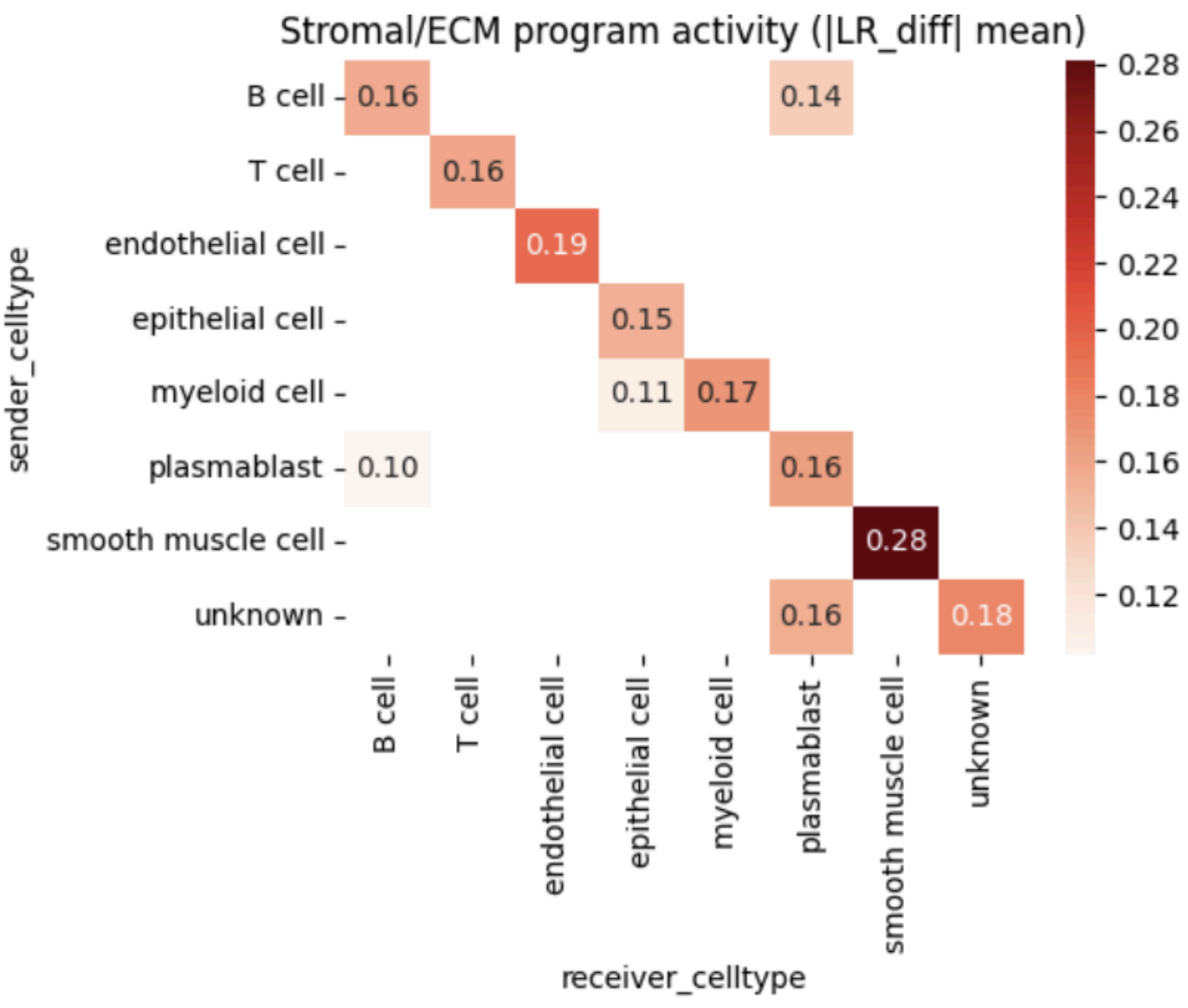
(Method 2A) VS (Method 2B)



Without Spatial

Observation: Smooth Muscle Autocrine signal is dominant and spatially stable (0.29 → 0.28).

Bio-Inference: Identifies the "Desmoplastic Reaction", dense, fibrotic bands of Myofibroblasts (CAFs) that structurally fortify the tumor.



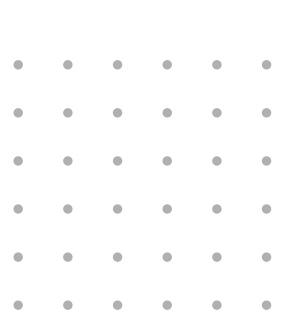
With Spatial

Observation: Epithelial → Myeloid interaction becomes visible in the Spatial model (0.11).

Bio-Inference: Highlights the possibility that the specific recruitment of Macrophages by Tumor cells at the invasive edge to degrade the matrix, a feature missed by non-spatial analysis.

Observation: Endothelial → Plasmablast links are removed.

Bio-Inference: Accurately separates vascular basement membrane signaling from the interstitial stromal matrix.



Conclusion

Framework Overview : Integrated COMMOT-derived LR matrices with an adapted SCDisInFact model under spatial and non-spatial settings.

Non-Spatial Findings

Adapted scDisInFact model uncovered coherent LR communication programs, including: *EMT/Invasion, Stromal–ECM, Immune-regulatory, and Growth-factor pathways*. Identified broad cancer-associated increases in:

- *Stromal remodelling*
- *Invasion-linked signalling*

Spatial Model Enhancements

Adding spatial coordinates produced sharply localised signalling hubs. Improved sender–receiver specificity and revealed microenvironmental niches involving:

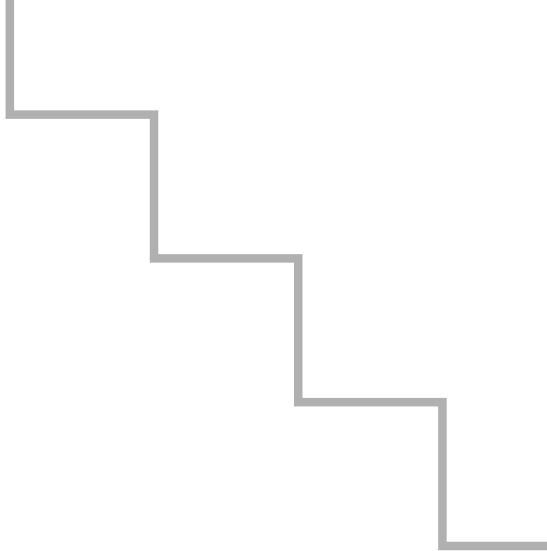
- Epithelial–myeloid
- Epithelial–stromal
- Immune–stromal interactions

Key Differential Pathways (Consistently Cancer-Enriched)

- MDK → SDC/NCL
- ANGPTL4 → syndecans
- CXCL12 → CXCR4
- PROS1/GAS6 → AXL
- PDGF / VEGF signalling
- Reflect biological processes: Matrix remodelling, Invasion, Immune modulation, Angiogenesis.

Provides a generalizable framework for studying multicellular signalling in complex tissues.

Establishes a foundation for applications to perturbation experiments and multimodal spatial datasets.



THANK YOU

TEAM-7

Vaneesha S Kumar

Nischay Patel

Zartaf Raza

Anshul Ranka

